Hardin's Arrangements in Which Every Number Is a Sum of Others: Formulas and Sumsets

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Abstract

In 2011 Hardin contributed to the OEIS a family of sequences counting arrangements of residues modulo a fixed integer in which every entry is the sum of two, or of three, of the other entries; most of them carry only empirical formulas or conjectures. We show that whether such an arrangement qualifies depends only on how often each residue occurs, counted up to a small threshold, which turns the enumeration into a finite sum of exponential generating functions. Hence every sequence of the family is eventually a sum of polynomials times exponentials (for ordered arrangements) or a polynomial (for nondecreasing ones). We give explicit formulas for all twenty-eight sequences of the family, prove the empirical formulas and conjectures recorded in the OEIS that were still open, and show that three of the conjectured recurrences hold only from one step later than stated. We also show that an exponential with base smaller than the modulus can occur only if some set of that many residues fails to produce every residue as a sum of two (or three) of its elements, and we conjecture that this condition is also sufficient; the conjecture holds in all cases computed.

1 Introduction

Fix integers $q \geq 2$ and $s \geq 2$. In January 2011 Hardin contributed to the On-Line Encyclopedia of Integer Sequences [6] a family of sequences with titles such as "Number of arrangements of $n+2$ numbers in 0..6 with each number being the sum mod 7 of two others" (A183881), for sums of two or of three others and $3 \leq q \leq 9$, together with the corresponding "nondecreasing arrangements" and four tables collecting them; see Table 1.

For many of these sequences the OEIS records formulas only as empirical (found by fitting the terms) or as conjectures: closed forms by Kotesovec for $q \leq 5$ ($s = 2$) and $q \leq 4$ ($s = 3$), polynomial formulas by Hardin for all nondecreasing arrangements, and conjectured generating functions and recurrences by Barker. Israel proved the conjectures recorded for A183877, and he proved that every column of A183912 (nondecreasing arrangements, $s = 2$)

family	s	$q = 3, 4, \ldots, 9$	table
arrangements	2	A183877–A183883	A183884
arrangements	3	A183885–A183891	A183892
nondecreasing arrangements	3	A183897–A183903	A183904
nondecreasing arrangements	2	A183905–A183911	A183912

Table 1: The thirty-two sequences of Hardin's family.

is eventually a polynomial [4]. Eight sequences of the family have no formula at all, and neither do the four tables.

We prove that all these sequences are of a very simple type. The key observation (Lemma 3) is that whether a word qualifies depends only on how many times each residue occurs, counted up to the threshold $s + 1$. Counting words with a prescribed number of occurrences of each letter is classical, and summing over the finitely many admissible patterns gives the results: the number of ordered arrangements is eventually a sum of polynomials times exponentials k^m (Theorem 5), and the number of nondecreasing arrangements is eventually a polynomial in m (Theorem 6). The finite sum is evaluated by computer; this gives explicit formulas for all twenty-eight sequences (Section 5), settles every empirical formula and conjecture recorded in the OEIS for the family (all are true, except that three recurrences hold only from one step later than stated), and supplies formulas for the sequences that had none (Section 6).

The exponentials that occur are governed by additive structure (Section 7): k^m with $k < q$ can occur only if some set of k residues does not produce every residue as a sum of s of its elements (Theorem 7); in particular k^m never occurs for $q/2 < k < q$. We conjecture that this condition is also sufficient (Conjecture 12) and verify it for $q \leq 14$ ($s = 2$) and $q \leq 11$ ($s = 3$).

Israel's argument for A183912 also records each residue only up to three occurrences; it is the case $s = 2$ of Lemma 3 for nondecreasing words, and our Theorem 6 extends his result. Apart from these results in the OEIS we have not found earlier work on these counts. In additive combinatorics a subset of an abelian group is called *sum-full* if every element of it is a sum of two other elements, possibly equal to each other; Lev, Nagy, and Pach [5] answered a question of Zaimi on such sets. For $s = 2$ the words counted here are the multiset analogue of sum-full subsets of $\mathbb{Z}_q$, with the two summands taken at distinct positions. Chung, Folkman, and Graham [1] studied a different condition on sequences modulo n, namely that every residue is represented the same number of times as a sum of two entries.

2 Definitions

Let $\mathbb{Z}_q = \{0, 1, \ldots, q - 1\}$ denote the integers modulo q.

Definition 1. A word $w = w_1 w_2 \cdots w_m$ over $\mathbb{Z}_q$ is *s-covered* if for every position i there are s distinct positions $j_1, \ldots, j_s$, all different from i, such that $w_{j_1} + \cdots + w_{j_s} \equiv w_i \pmod q$.

Example 2. Over $\mathbb{Z}_7$ the word 112244 is 2-covered: $2 = 1+1$, $4 = 2+2$, and $1 \equiv 4+4 \pmod 7$, each time with two positions different from the one being covered. The word 001 is not 2-covered, since the only sum of two other entries available for the last position is $0+0=0$.

Let $W_{q,s}(m)$ denote the number of s-covered words of length m, and let $N_{q,s}(m)$ denote the number of s-covered words of length m that are nondecreasing (equivalently, s-covered multisets of size m). Hardin's sequences are $a(n) = W_{q,s}(n+s)$ and $a(n) = N_{q,s}(n+s)$ with offset 1. We checked this reading of the definition ("s distinct other positions") by brute force against the terms in the OEIS for small m.

For a word w and a residue v let $c_v(w)$ be the number of positions i with $w_i = v$. The *profile* of w is the map $\pi_w \colon \mathbb{Z}_q \to \{0, 1, \ldots, s+1\}$, $\pi_w(v) = \min(c_v(w), s+1)$. We call v *frequent* in w if $\pi_w(v) = s+1$.

3 Profiles

Lemma 3. *Whether a word is s-covered depends only on its profile.*

Proof. Let $w_i = v$. Position i is covered if and only if there is a multiset $\{a_1, \ldots, a_s\}$ of residues with $a_1 + \cdots + a_s \equiv v \pmod q$ such that for every residue a the multiplicity μ_a of a in the multiset satisfies $\mu_a \le c_a(w) - [a = v]$, where $[\cdot]$ is the Iverson bracket. Since $\mu_a + [a = v] \le s+1$, this inequality holds if and only if it holds with $c_a(w)$ replaced by $\min(c_a(w), s+1)$. The condition does not depend on i beyond the letter w_i. □

By Lemma 3 we may speak of *valid* profiles. Let $\mathcal{V}_{q,s}$ be the set of valid profiles; it is a subset of the $(s+2)^q$ maps $\mathbb{Z}_q \to \{0, \ldots, s+1\}$.

Lemma 4. *Let π be a profile. Put $f_c(t) = t^c/c!$ for $0 \le c \le s$ and $f_{s+1}(t) = e^t - p_s(t)$, where $p_s(t) = \sum_{j=0}^{s} t^j/j!$. Then the number of words of length m with profile π is*

$$m!\,[t^m] \prod_{v \in \mathbb{Z}_q} f_{\pi(v)}(t),$$

and the number of nondecreasing words of length m with profile π is $[y^m] \prod_{v \in \mathbb{Z}_q} g_{\pi(v)}(y)$, where $g_c(y) = y^c$ for $c \le s$ and $g_{s+1}(y) = y^{s+1}/(1-y)$.

Proof. A word is determined by the sets of positions of its letters, so the exponential generating function of words with c_v letters v, summed over the allowed values of each c_v, is the product of the exponential generating functions $\sum_c t^c/c!$ over the allowed values [3, Ch. II]. A nondecreasing word is determined by the counts alone. □

4 Main results

Theorem 5. *Let $q \geq 2$ and $s \geq 2$. There are polynomials $P_1, \ldots, P_q$ and an integer m_0 such that*

$$W_{q,s}(m) = \sum_{k=1}^{q} P_k(m)\, k^m \qquad (m > m_0).$$

Moreover $P_q = 1$ and $\deg P_k \leq s(q-k)$ whenever $P_k \neq 0$. Consequently $\sum_{m\geq 0} W_{q,s}(m)x^m$ is a rational function whose denominator divides $\prod(1-kx)^{\deg P_k+1}$, and $W_{q,s}(m)$ satisfies the linear recurrence with constant coefficients and characteristic polynomial $\prod(x-k)^{\deg P_k+1}$ for all sufficiently large m, both products taken over the k with $P_k \neq 0$.

Proof. By Lemmas 3 and 4, $W_{q,s}(m) = m!\,[t^m]F(t)$ with $F(t) = \sum_{\pi\in\mathcal{V}_{q,s}} \prod_v f_{\pi(v)}(t)$. A profile with r frequent residues contributes $(e^t - p_s(t))^r$ times a monomial of degree at most $s(q-r)$; expanding gives a finite sum $F(t) = \sum_{a,k} c_{a,k}\, t^a e^{kt}$ with $0 \leq k \leq q$, and the terms with a given $k \geq 1$ have $a \leq s(q-k)$. Since $m!\,[t^m]\, t^a e^{kt} = m(m-1)\cdots(m-a+1)\, k^{m-a}$, the terms with $k \geq 1$ give $P_k(m)k^m$, and the terms with $k = 0$ vanish for $m > m_0$, where m_0 is the largest a with $c_{a,0} \neq 0$. Only the profile in which every residue is frequent contributes to e^{qt}; it is valid, because every residue is a sum of s residues, which gives $P_q = 1$. The statements on the generating function and the recurrence follow from the form of $W_{q,s}(m)$. □

Theorem 6. *Let $q \geq 2$ and $s \geq 2$. There is an m_0 such that $N_{q,s}(m)$ is a polynomial in m of degree $q-1$ with leading coefficient $1/(q-1)!$ for $m > m_0$. Its generating function is rational with denominator dividing $(1-y)^q$.*

Proof. By Lemma 4 the generating function is $\sum_{\pi\in\mathcal{V}_{q,s}} y^{e(\pi)}/(1-y)^{r(\pi)}$, where $r(\pi)$ is the number of frequent residues and $e(\pi) = \sum_v \pi(v)$. For $r \geq 1$ the coefficient of y^m in $y^e/(1-y)^r$ is $\binom{m-e+r-1}{r-1}$, a polynomial in m of degree $r-1$ for $m \geq e-r+1$; for $r = 0$ it vanishes for $m > e$. Only the profile in which all residues are frequent has $r = q$. □

For $s = 2$, the fact that $N_{q,2}(m)$ is eventually a polynomial in m is due to Israel [4], who obtained it from a transfer matrix on the same states; the degree and the leading coefficient are new.

For a set $B \subseteq \mathbb{Z}_q$ let $sB = \{b_1 + \cdots + b_s : b_1, \ldots, b_s \in B\}$ denote its s-fold sumset (repetitions allowed).

Theorem 7. *Let $q \geq 2$, $s \geq 2$, and $1 \leq k < q$. If $sB = \mathbb{Z}_q$ for every set $B \subseteq \mathbb{Z}_q$ of k residues, then $P_k = 0$ in Theorem 5.*

Proof. Write $f_{s+1} = e^t - p_s$ and expand $F(t)$. A term containing e^{kt} arises from a valid profile together with a set B of k of its frequent residues, which take the factor e^t; every other residue $u \notin B$ takes the weight $t^j/j!$ if $\pi(u) = j \leq s$, and the weight $-p_s(t)$ if u is frequent. Fix B with $sB = \mathbb{Z}_q$ and suppose that all residues of B are frequent. Every residue v can be written as $b_1 + \cdots + b_s$ with $b_i \in B$; a residue occurs at most s times among

the summands, and at most $s+1$ times if the covered position is counted, so the frequent residues supply enough distinct positions. Hence every profile in which B is frequent is valid, whatever the states of the residues outside B, and the total weight of these states is $\prod_{u\notin B}\bigl(\sum_{j=0}^{s} t^j/j! - p_s(t)\bigr) = 0$, since $k < q$. Summing over B gives $P_k = 0$. □

Corollary 8. *Let $q \geq 2$ and $s \geq 2$.*

1. *$P_k = 0$ for $q/2 < k < q$.*
2. *If q is prime, then $P_k = 0$ for $(q-1)/s + 1 \leq k < q$.*

Proof. If $|B| = k > q/2$, then for every residue v the sets B and $v - B$ meet, so $v \in B + B$. Hence $B + B = \mathbb{Z}_q$, and $sB = \mathbb{Z}_q$ for every $s \geq 2$. If q is prime, the Cauchy–Davenport theorem [2], applied $s-1$ times, gives $|sB| \geq \min(q, s(k-1)+1)$, which equals q for $k \geq (q-1)/s + 1$. In both cases Theorem 7 applies. □

5 Explicit formulas

In this section $a(n)$ denotes the corresponding OEIS sequence (offset 1), that is, $a(n) = W_{q,s}(n+s)$ or $a(n) = N_{q,s}(n+s)$. For each sequence we give a number n_0 and an expression; the statement is that $a(n)$ equals the expression for all $n \geq n_0$ and differs from it for $1 \leq n < n_0$.

Theorem 9. *For the arrangements with $s = 2$, the following holds.*
A183877 *($q = 3$, $n_0 = 2$):*

$$a(n) = 3^{n+2} - 2(n+3)^2.$$

A183878 *($q = 4$, $n_0 = 3$):*

$$a(n) = 4^{n+2} - (2n+7)\cdot 2^{n+2} - (2n^3 + 9n^2 + 10n - 3).$$

A183879 *($q = 5$, $n_0 = 2$):*

$$\begin{aligned} a(n) = {} & 5^{n+2} \\ & - (5n^2 + 35n + 62)\cdot 2^{n+1} \\ & + 6n^3 + 46n^2 + 128n + 128. \end{aligned}$$

A183880 *($q = 6$, $n_0 = 3$):*

$$\begin{aligned} a(n) = {} & 6^{n+2} \\ & - (4n+18)\cdot 3^{n+2} \\ & - (5n^3 + 35n^2 + 6n - 256)\cdot 2^{n-1} \\ & + 2n^4 + 23n^3 + 85n^2 + 117n + 29. \end{aligned}$$

A183881 ($q = 7$, $n_0 = 4$):

$$\begin{aligned} a(n) = 7^{n+2} \\ &- (28n^2 + 252n + 554) \cdot 3^n \\ &+ (9n^3 + 201n^2 + 1044n + 1668) \cdot 2^{n-1} \\ &+ 6n^4 + 19n^3 - 45n^2 - 274n - 342. \end{aligned}$$

A183882 ($q = 8$, $n_0 = 4$):

$$\begin{aligned} a(n) = 8^{n+2} \\ &- (8n + 43) \cdot 4^{n+2} \\ &- (16n^3 + 156n^2 - 76n - 2646) \cdot 3^{n-1} \\ &+ (2n^4 + 63n^3 + 435n^2 + 894n - 128) \cdot 2^{n-1} \\ &+ 7n^4 - 28n^3 - 256n^2 - 496n - 263. \end{aligned}$$

A183883 ($q = 9$, $n_0 = 4$):

$$\begin{aligned} a(n) = 9^{n+2} \\ &- (70n^2 + 786n + 2076) \cdot 4^n \\ &+ (11n^3 + 753n^2 + 5782n + 12420) \cdot 3^{n-1} \\ &+ (9n^4 + 78n^3 - 219n^2 - 2700n - 5268) \cdot 2^{n-1} \\ &- (3n^5 + 57n^4 + 245n^3 + 373n^2 - 8n - 438). \end{aligned}$$

Theorem 10. *For the arrangements with $s = 3$, the following holds.*

A183885 ($q = 3$, $n_0 = 3$):

$$a(n) = 3^{n+3} - 2(n+4)^2.$$

A183886 ($q = 4$, $n_0 = 2$):

$$a(n) = 4^{n+3} - (n+3) \cdot 2^{n+4} - (14n^3 + 102n^2 + 250n + 216)/3.$$

A183887 ($q = 5$, $n_0 = 4$):

$$\begin{aligned} a(n) = 5^{n+3} \\ &- (3n + 13) \cdot 2^{n+3} \\ &- (17n^4 + 182n^3 + 733n^2 + 1312n + 864)/3. \end{aligned}$$

A183888 ($q = 6$, $n_0 = 5$):

$$\begin{aligned} a(n) = 6^{n+3} \\ &- (2n + 6) \cdot 3^{n+3} \\ &- (n+4)^2 \cdot 2^{n+5} \\ &- (5n^5 + 99n^4 + 509n^3 + 693n^2 - 1090n - 2592)/6. \end{aligned}$$

A183889 ($q = 7,\ n_0 = 4$):

$$\begin{aligned} a(n) = 7^{n+3} \\ &- (25n^3 + 330n^2 + 1451n + 2154) \cdot 2^n \\ &- (13n^5 + 107n^4 + 246n^3 - 236n^2 - 1762n - 2046). \end{aligned}$$

A183890 ($q = 8,\ n_0 = 5$):

$$\begin{aligned} a(n) = 8^{n+3} \\ &- (2n + 6) \cdot 4^{n+3} \\ &- (8n + 32) \cdot 3^{n+3} \\ &- (83n^4 + 1874n^3 + 11761n^2 + 27634n + 19584) \cdot 2^{n-2}/3 \\ &- (50n^5 + 93n^4 - 1662n^3 - 8595n^2 - 15992n - 10848)/3. \end{aligned}$$

A183891 ($q = 9,\ n_0 = 6$):

$$\begin{aligned} a(n) = 9^{n+3} \\ &- (17n^2 + 193n + 522) \cdot 3^{n+2} \\ &- (17n^5 + 641n^4 + 7661n^3 + 32335n^2 + 41330n - 17184) \cdot 2^{n-4} \\ &- (16n^6 + 142n^5 - 95n^4 - 4054n^3 - 14569n^2 - 21112n - 11024)/2. \end{aligned}$$

Theorem 11. *For the nondecreasing arrangements, the following holds.*

A183905 ($q = 3,\ s = 2,\ n_0 = 2$):

$$a(n) = \frac{n^2 + 7n - 10}{2}.$$

A183906 ($q = 4,\ s = 2,\ n_0 = 3$):

$$a(n) = \frac{n^3 + 12n^2 + 5n - 48}{6}.$$

A183907 ($q = 5,\ s = 2,\ n_0 = 2$):

$$a(n) = \frac{n^4 + 18n^3 + 119n^2 - 570n + 552}{24}.$$

A183908 ($q = 6,\ s = 2,\ n_0 = 3$):

$$a(n) = \frac{n^5 + 25n^4 + 245n^3 - 145n^2 - 2766n + 5160}{120}.$$

A183909 ($q = 7,\ s = 2,\ n_0 = 4$):

$$a(n) = \frac{n^6 + 33n^5 + 445n^4 + 3135n^3 - 26006n^2 + 38232n + 8640}{720}.$$

A183910 ($q=8,\ s=2,\ n_0=4$):

$$a(n)=\frac{1}{5040}\big(n^7+42n^6+742n^5+7140n^4-9191n^3-223902n^2 +714048n-418320\big).$$

A183911 ($q=9,\ s=2,\ n_0=4$):

$$a(n)=\frac{1}{40320}\big(n^8+52n^7+1162n^6+14560n^5+111769n^4 -1283492n^3+2144988n^2+4978320n-11975040\big).$$

A183897 ($q=3,\ s=3,\ n_0=3$):

$$a(n)=\frac{n^2+9n-2}{2}.$$

A183898 ($q=4,\ s=3,\ n_0=2$):

$$a(n)=\frac{n^3+15n^2+50n-84}{6}.$$

A183899 ($q=5,\ s=3,\ n_0=4$):

$$a(n)=\frac{n^4+22n^3+179n^2+398n-1944}{24}.$$

A183900 ($q=6,\ s=3,\ n_0=5$):

$$a(n)=\frac{n^5+30n^4+355n^3+1710n^2-2096n-6000}{120}.$$

A183901 ($q=7,\ s=3,\ n_0=4$):

$$a(n)=\frac{n^6+39n^5+625n^4+5265n^3+24574n^2-106104n-31680}{720}.$$

A183902 ($q=8,\ s=3,\ n_0=5$):

$$a(n)=\frac{1}{5040}\big(n^7+49n^6+1015n^5+11515n^4+70504n^3 +144676n^2-1669200n+1572480\big).$$

A183903 ($q=9,\ s=3,\ n_0=6$):

$$a(n)=\frac{1}{40320}\big(n^8+60n^7+1554n^6+22680n^5+203889n^4 +1155420n^3+56636n^2-28575600n+32296320\big).$$

Proof of Theorems 9–11. By the proofs of Theorems 5 and 6, each sequence is given by an explicit finite sum over the valid profiles. We computed the set $\mathcal{V}_{q,s}$ and expanded these sums in exact rational arithmetic (Section 8); this gives the expressions above, valid for $n+s>m_0$. For the finitely many remaining n the expressions were compared with the exact counts, which gives n_0. □

6 Consequences for the OEIS

Theorems 9–11 settle the formulas recorded for the family in the OEIS.

- The empirical closed forms of Kotesovec for A183877, A183878, A183879, A183885, and A183886, the empirical polynomial formulas of Hardin for A183897–A183903 and A183905–A183911, and the conjectured generating functions and recurrences of Barker for these sequences (except A183902 and A183903, which have none) are all true, with the ranges of validity stated there, with three exceptions given in the next item. For A183877 this was already proved by Israel in the OEIS entry. The description of A183905 as triangular numbers minus 11 is also correct: $a(n) = \binom{n+4}{2} - 11$ for $n \geq 2$.

- The recurrences of Barker for A183908, A183910, and A183911, stated for $n > 7$, $n > 10$, and $n > 11$, hold exactly for $n > 8$, $n > 11$, and $n > 12$. This agrees with Theorem 11: a polynomial formula valid for $n \geq n_0$ of degree $d-1$ gives the recurrence of order d for $n \geq n_0 + d$.

- The conjectured closed form and generating function of A183880 are true.

- A183881–A183883 and A183887–A183891 had no formula; Theorems 9 and 10 supply one.

- Every column of the tables A183884, A183892, A183904, and A183912 is described by Theorem 5 or Theorem 6, including the columns for $q = 2$. For these the same computation gives $W_{2,2}(m) = 2^m - m - 1$ and $W_{2,3}(m) = 2^m - 2m$ for $m \geq s+1$ (that is, for all $n \geq 1$ in the OEIS indexing), which proves that the first columns of A183884 and A183892 are A000295 and A005803, as stated in the OEIS.

7 Which exponentials occur

Table 2 lists, for $s \in \{2, 3\}$, the values k with $P_k \neq 0$ in Theorem 5 together with $\deg P_k$.

Let

$$K(q, s) = \max\{|B| : B \subseteq \mathbb{Z}_q,\ sB \neq \mathbb{Z}_q\}.$$

Every subset of a set B with $sB \neq \mathbb{Z}_q$ has the same property, so sets of every size $k \leq K(q, s)$ fail to cover $\mathbb{Z}_q$, and by Theorem 7 we have $P_k = 0$ for $K(q, s) < k < q$. For instance $K(q, 2) = \lfloor q/2 \rfloor$, as Corollary 8 and the set $\{0, 1, \ldots, \lfloor q/2 \rfloor - 1\}$ show; $K(q, 3) = q/2$ for even q (take the even residues); and for prime q the Cauchy–Davenport bound is attained by arithmetic progressions, so $K(q, 3)$ is the largest k with $3k - 2 < q$. Table 2 shows that the converse of Theorem 7 holds in every computed case.

Conjecture 12. For $q \geq 2$, $s \geq 2$, and $1 \leq k < q$, we have $P_k \neq 0$ if and only if $k \leq K(q, s)$. Equivalently, the exponentials in $W_{q,s}(m)$ are exactly q^m and k^m for $1 \leq k \leq K(q, s)$.

s	q	exponentials (degree of P_k)
2	3	3^m (0), 1 (2)
2	4	4^m (0), 2^m (1), 1 (3)
2	5	5^m (0), 2^m (2), 1 (3)
2	6	6^m (0), 3^m (1), 2^m (3), 1 (4)
2	7	7^m (0), 3^m (2), 2^m (3), 1 (4)
2	8	8^m (0), 4^m (1), 3^m (3), 2^m (4), 1 (4)
2	9	9^m (0), 4^m (2), 3^m (3), 2^m (4), 1 (5)
2	10	10^m (0), 5^m (1), 4^m (3), 3^m (4), 2^m (5), 1 (6)
2	11	11^m (0), 5^m (2), 4^m (3), 3^m (4), 2^m (5), 1 (5)
2	12	12^m (0), 6^m (1), 5^m (3), 4^m (4), 3^m (5), 2^m (6), 1 (6)
2	13	13^m (0), 6^m (2), 5^m (3), 4^m (4), 3^m (5), 2^m (5), 1 (5)
2	14	14^m (0), 7^m (1), 6^m (3), 5^m (4), 4^m (5), 3^m (6), 2^m (6), 1 (7)
3	3	3^m (0), 1 (2)
3	4	4^m (0), 2^m (1), 1 (3)
3	5	5^m (0), 2^m (1), 1 (4)
3	6	6^m (0), 3^m (1), 2^m (2), 1 (5)
3	7	7^m (0), 2^m (3), 1 (5)
3	8	8^m (0), 4^m (1), 3^m (1), 2^m (4), 1 (5)
3	9	9^m (0), 3^m (2), 2^m (5), 1 (6)
3	10	10^m (0), 5^m (1), 4^m (1), 3^m (3), 2^m (5), 1 (6)
3	11	11^m (0), 4^m (1), 3^m (4), 2^m (5), 1 (6)

Table 2: Exponentials k^m in $W_{q,s}(m)$.

We verified Conjecture 12 for $q \leq 14$ when $s = 2$ and for $q \leq 11$ when $s = 3$; the cases $q = 13$ and $q = 14$ with $s = 2$ were computed after the pattern had been noticed in the smaller cases. The proved part, Corollary 8, already explains some absences below $q/2$: for instance 3^m does not occur for $q = 7$ and $s = 3$, although $3 < 7/2$, because every set of three residues modulo 7 produces all residues as sums of three of its elements.

8 Computation

The set $\mathcal{V}_{q,s}$ was obtained by testing all $(s+2)^q$ profiles with Lemma 3; for $q \leq 9$ this takes about a second per pair (q, s) in C++, 40 seconds for $q = 11$, $s = 3$, and about three minutes for $q = 14$, $s = 2$. For instance $|\mathcal{V}_{7,2}| = 13234$ and $|\mathcal{V}_{5,3}| = 2640$. Since the product in Lemma 4 depends only on how many residues fall into each class, it suffices to record the histogram of the valid profiles by the numbers of residues with each value of π. The sums were then expanded in exact rational arithmetic. As checks, the resulting counts agree with all terms of the OEIS entries (up to $n = 200$), each formula in Section 5 was evaluated from

its printed form and compared with the exact counts for at least sixty consecutive values of n, and the counts for small m were confirmed by direct enumeration of words. The programs and the computed data are available at github.com/mindscanlabs/NAI-track-record.

9 Acknowledgments

The closed forms for A183880, A183881, and A183887 were first found from linear recurrences guessed by NAI (No-Ansatz Inference), the author's program for finding formulas that fit given data without fixing their form in advance; it does not use language models. The recurrences were guessed from initial terms and checked against further terms. Claude (Anthropic), an AI assistant, carried out the computations of this paper, found the profile argument, and drafted the proofs and the text. The author checked the results, takes responsibility for the content, and thanks the editors of the OEIS for their comments on related submissions.

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